

Real-Time Speech Translation Using LSTM

V.Gowthami

20951A0455

Real-Time Speech Translation Using LSTM

*A project report
submitted in partial fulfilment of the
requirements for the award of the degree of*

**Bachelor of Technology
in
Electronics and Communication Engineering**

by

**V. Gowthami
20951A0455**

**Under the esteemed guidance of
Dr Surekha Reddy Bandela,
Assistant Professor**



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

INSTITUTE OF AERONAUTICAL ENGINEERING

(Autonomous)

Dundigal, Hyderabad-500043, Telangana

April, 2024

© V. Gowthami. All rights reserved

DECLARATION

I certify that

- a. The work contained in this report is original and has been done by me under the guidance of my supervisor(s).
- b. The work has not been submitted to any other Institute for any degree or diploma.
- c. I have followed the guidelines provided by the Institute in preparing the report.
- d. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- e. Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Place:

Date:

Signature:

Roll Number:

CERTIFICATE

This is to certify that the project report entitled **Real-Time Speech Translation Using LSTM** submitted by **Miss.V.Gowthami** to the Institute of Aeronautical Engineering, Hyderabad in partial fulfillment of the requirements for the award of the Degree Bachelor of Technology in **Electronics and Communication Engineering** is a bonafide record of work carried out by him/her under my/our guidance and supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute for the award of any Degree.

Supervisor

Head of the Department

Date:

APPROVAL SHEET

This project report entitled **Real-Time Speech Translation Using LSTM** by **Electronics and Communication Engineering** is approved for the award of the Degree Bachelor of Technology in **Electronics and Communication Engineering**

Examiner:

Supervisor:

Principal

Dr L.V. Narasimha Prasad

Date:

Place:

ACKNOWLEDGEMENT

I would like to take this opportunity to extend our gratitude to all those who assisted, encouraged and extended their cooperation in various ways during our project. It is my pleasure to recognize those individuals who are responsible for foreseeing the successful completion of our work.

I want to thank **Dr. P Munaswamy (ECE,HOD)** and express my gratitude with great admiration and respect to my project guide **Dr. Surekha Reddy Bandela (Assistant Professor, Department of Electronics and Communication Engineering)**, for their valuable advice and help throughout the development of this project by providing with required information, without whose guidance, cooperation and encouragement, this project couldn't have been materialized.

Last but not least, I would like to thank my principal, **Dr L.V. Narasimha Prasad**, for extending his help in all circumstances

With Gratitude,
V Gowthami
20951A0455

ABSTRACT

For tackling the challenge of translating spoken words which involves converting speech from one language to another in time. In today's interconnected world facilitating communication, between individuals of linguistic backgrounds is a complex yet crucial task. To overcome this obstacle we have designed an internet based platform that utilizes written text as a medium to facilitate speech translation. Our approach draws on experiments involving translating voice messages offline, converting spoken words to text online and transforming spoken words into spectrograms offline as the foundation of our strategy. Furthermore, our thesis compares hidden units with conventional text-based approaches and thoroughly examines their potential and usefulness as intermediate representations for translation tasks. We investigate if hidden units can provide more accurate and insightful voice representations to support the translation process. We evaluate our suggested system's efficacy through experiments and contrast its results with those of other methods. Metrics like average delay and BLEU scores for the text generated by our online model are used to evaluate it. We provide a thorough analysis of the findings, emphasising the benefits and limitations of our approach. Additionally, we discuss the technical difficulties that come with the job, like managing accents, dialects, and speaker variances, and we offer solutions for these issues.

Keywords: Speech-to-Text translation , Latency, BLUE score , Text-to-Speech translation, Word Accuracy, Simultaneous Speech.

CONTENTS

TITLE	PAGE NO.
Title Page	i
Inside cover page	ii
Declaration	iii
Certificate by the Supervisor	iv
Approval Sheet	v
Acknowledgement	vi
Abstract	vii
Contents	viii
List of Tables	x
List of Figures	xi
List of Abbreviations	xii
CHAPTER 1: Introduction	1
1.1 Introduction to Speech Translation	1
1.2 Significance of Speech Translation	2
1.3 Scope of Speech Translation in Real-time	3
1.4 Thesis Organization	4
CHAPTER 2: Literature Survey	6
2.1 Overview of Speech to Speech Translation	6
2.2 Speech Encoders	7
2.3 Speech to Unit Translations	8
2.4 End-to-End/Direct Speech to Speech Translations	9
2.5 Attention Mechanisms	10
2.5.1 Monotonic Attention	11
2.5.2 Monotonic Chunkwise Attention	11
2.5.3 Monotonic Infinite Lookback Attention	13

2.5.4 Monotonic Multihead Attention	14
2.6 Simultaneous Speech to Text Translation	15
2.7 Automatic Speech Recognition	16
CHAPTER 3: Methodology	19
3.1 System Architecture	19
3.2 Data Collection	20
3.2.1 CoVoST and CoVoST 2	20
3.2.2 Common Voice Mozilla	21
3.3 Algorithms	21
3.3.1 Long Sort Term Memory (LSTM) Algorithm	22
3.3.2 Large Language Model (LLM)	23
CHAPTER 4: Implementation	26
4.1 Development Environment	26
4.1.1 SimulST Component	28
4.1.2 TTS Component	29
4.1.3 Pipeline System Design	29
4.2 BLEU Score evaluation	31
4.3 Policy Comparisons	31
CHAPTER 5: Results and Discussion	33
5.1 Simulation Output	33
5.1.1 BLEU Score Comparison	33
5.1.2 Word Accuracy	35
CHAPTER 6: Conclusion and Future Scope	36
6.1 Conclusion	36
6.2 Future scope	36

LIST OF TABLES

Table no.	Title	Page no.
5.1	BLEU Score comparison	33

LIST OF FIGURES

Figure no.	Title	Page no.
1.1	Schematic of Cascaded Speech to Speech Translation System	1
1.2	Translation process at a particular time	2
2.1	Prediction of hidden cluster assignments of masked frames	8
2.2	Illustration of textless S2ST model	9
2.3	Sample mel-spectrograms	10
2.4	Schematics of attention mechanisms	12
2.5	Schematics of the infinite lookback attention mechanism	13
2.6	Monotonic Attention(left) versus Monotonic Multihead Attention(right)	15
2.7	Simul-ST architecture with pre-decision module	16
2.8	Sequence-to-sequence transformer model	17
3.1	Schematic of Cascaded Simultaneous S2ST system	19
3.2	LSTM Algorithm flow chart	22
3.3	LLM Architecture	23
4.1	Proposed Cascaded S2ST System using OpenAI and ElevenLabs	26
4.2	Development of Simultaneous Speech Translation system	27

5.1	Visual Output of Cascaded S2ST System using OpenAI	34
5.2	Results of Word Accuracy	35

LIST OF ABBREVIATIONS

S2ST	Speech to Speech Translation
ASR	Automatic Speech Recognition
MT	Machine Translation
TTS	Text-to-Speech synthesis
LSTM	Long Short Term Memory
HuBERT	Hidden - Unit Bidirectional Encoder Representations from Transformers
S2UT	Speech - to – Unit Translation model
RNN	Recurrent Neural Network
MoChA	Monotonic Chunkwise Attention
MILK	Monotonic Infinite Lookback attention
NMT	Neural Machine Translation
MMA	Monotonic Multihead Attention
Simul-ST	Simultaneous Speech Translation
SOTA	State-of-the-art
CoVoST	Common Voice Speech-to-text Translation
ST	Speech -to-Text model
Simul-S2ST	Simultaneous Speech-to-Speech Translation
API	Application Programming Interface
BLEU	BiLingual Evaluation Understudy
ALCA	Computation- Aware Average Lagging metric
NLTK	Natural Language Tool Kit
NLP	Natural Language Processing
WER	Word Error Rate

Chapter 1

Introduction

Speech translation, from one language, to another is widely embraced in the fields of machine learning and natural language processing showing advancements in times. This technology holds the promise of revolutionizing how people communicate on a scale. We're embarking on a mission; providing speech, to speech translation. This involves translating spoken words while the speaker is still talking from one language, such, as Hindi or Tamil to English.

1.1 Introduction to Speech Translation

Translation of recorded speech from one language to another without the requirement for real-time translation during discussions is known as offline speech-to-speech translation. Offline voice translation permits a lag between the source speech and the translated output, in contrast to real-time speech-to-speech translation, which seeks to deliver translation instantly throughout a discussion.

Offline speech-to-speech translation can be achieved through cascaded methodologies, wherein automatic speech recognition (ASR) first converts spoken words into text. Next, machine translation (MT) translates the text, followed by text-to-speech (TTS) synthesis, which reconverts the translated text into speech. In Figure 1.1, this cascaded network is illustrated. The main difference is that full voice recordings are analyzed, which frees up more time and processing power for the translation process and may lead to improved translation accuracy[1].

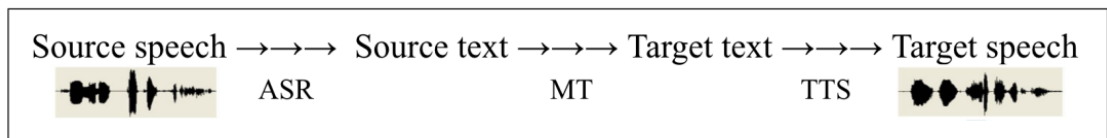


Figure 1.1: A schematic of Cascaded Speech to Speech Translation System.

We can observe a similar system in action through a simultaneous speech-to-text translation system demonstrated by Chen et al. (2021b), as depicted in Figure 1.2

illustrates the translation process during a specific time frame in the demo. This system accepts streaming speech input in one language and generates streaming text output in another language.

Previous efforts in speech-to-speech translation have relied on text as an intermediary representation. This process involves converting the source language speech into text, translating the text into the target language, and then converting that text back into speech using text-to-speech techniques. However, this method has limitations—it fails to preserve crucial speech elements such as emotion, hesitation, interruptions, and pauses, resulting in bland and robotic speech output. The challenge at hand is to execute this conversion process in real-time and with continuous streaming. The significance of this problem lies in the potential language barrier present between individuals speaking different languages. By reducing this barrier, we can facilitate easier communication among people, even if they are not fluent in each other's languages.

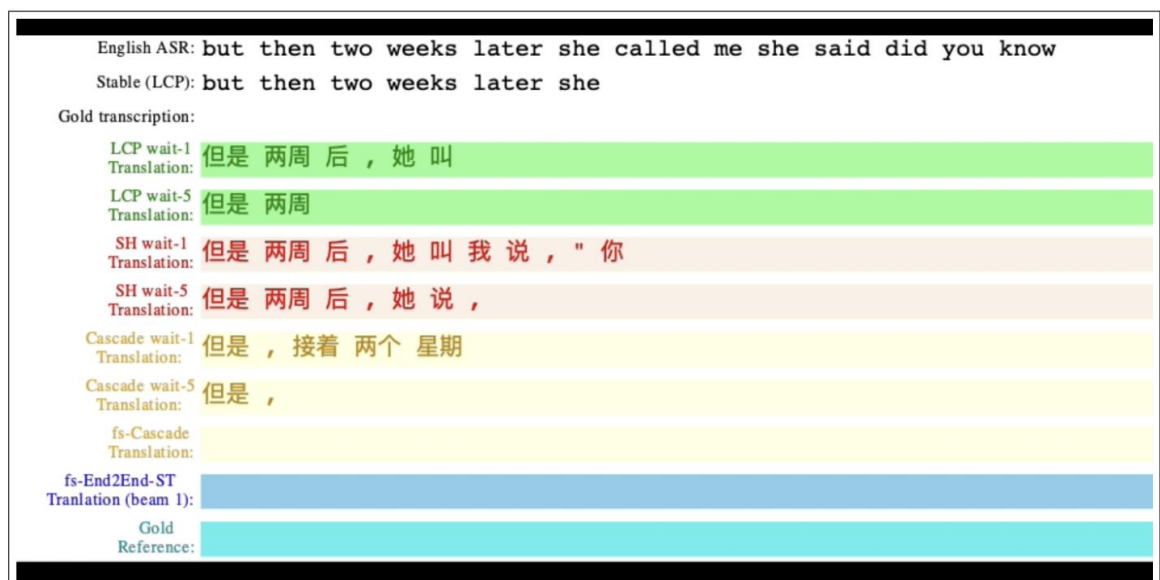


Figure 1.2: Translation process at a particular time step

1.2 Significance of Speech Translation

In numerous languages, a word that concludes a sentence in the original language could start the translated statement in the target language. Offline speech-to-speech translation has certain restrictions. The biggest disadvantage is the lag between the source voice and the translated output. This can be problematic in circumstances where

successful communication requires instantaneous translation. In situations like commercial discussions, customer service encounters, or emergency situations where real-time communication is crucial, offline speech-to-speech translation might not be appropriate.

A rapidly developing field of research is simultaneous speech-to-speech translation, which provides real-time translations of spoken language between two or more individuals. Simultaneous translation translates speech as it is spoken, as opposed to offline translation, which transcribes and translates the source language before producing the target language output. The procedure involves the real-time capture and processing of speech input, resulting in the prompt translation output that enables seamless communication between speakers of different languages.

The importance of speech-translation technology lies in its ability to bridge linguistic gaps in international commerce and cross-cultural interactions by facilitating communication between speakers of various languages. Speech translation would benefit humanity greatly from a scientific, cultural, and economic standpoint. S2ST, or automatic speech-to-speech translation, allows people to communicate in their native tongues, greatly lowering language barriers and bridging cultural gaps. Whether in a dialogue, conversation, or even a presentation, speech can help people understand each other's actions and thoughts. If the other person did not comprehend the language used for communication, the information would not be passed along effectively. Therefore, a system that can cross linguistic barriers is needed.

1.3 Scope of Speech Translation in Real-time

Simultaneous speech-to-speech translation poses several challenges. One major hurdle is the necessity for low-latency processing to ensure near real-time translation without significant delays that could disrupt conversation flow. Additionally, the accuracy of Automatic Speech Recognition (ASR) and Machine Translation (MT) technologies is crucial, as errors in recognition or translation can lead to misunderstandings. Another challenge lies in the system's ability to determine the optimal time to begin generating output confidently.

Subsequent developments in natural language processing and machine learning have propelled noteworthy breakthroughs in simultaneous speech-to-speech translation. Numerous strategies have been put forth, such as hybrid models, end-to-end models, and cascaded models. To improve translation accuracy and fluency, researchers have also looked into the application of deep neural networks, attention mechanisms, and reinforcement learning.

Simultaneous speech-to-speech translation is still a difficult research topic, despite recent advancements. More developments are required to improve the reliability and precision of translation systems so that they can be used in more situations. It is expected to play a crucial role in promoting cross-cultural communication and removing linguistic barriers as the sector continues to develop quickly.

Our goal is to lead the way in the development of simultaneous speech-to-speech translation (S2ST) by focusing on real-world uses such as live interpretation and cross-lingual voice communication. By using a two-step procedure, our methodology deviates from the limitations of direct end-to-end procedures. First, we convert source speech to target text using a SimulST component. The translated text is then sent into a text-to-speech (TTS) module, which generates target speech. With the help of our system, you may translate text from different languages into English with ease[2]. Our system's unique feature is its capacity to dynamically control the output delay using modifiable factors, such as four policies controlling when the output sequence is generated. System also provides modularity, making it easier to update or replace individual components without affecting the entire pipeline, which enhances the system's adaptability and maintainability.

1.4 Thesis Organization

The thesis is structured to provide a comprehensive exploration of the field. Beginning with an introduction that outlines the motivation, problem statement, and objectives, it proceeds to a thorough review of existing literature, highlighting challenges and recent advancements. The methodology section details the systematic approach to developing the translation system, from architecture design to evaluation metrics. Implementation discusses practical aspects such as data collection, model training, and integration. Experimental results present findings and comparisons with

existing systems, while discussion interprets results and suggests future research directions. The conclusion summarizes key contributions and outlines implications for the field. The concise organization ensures a clear progression from problem identification to solution development and evaluation, aiming to advance real-time speech translation technology.

Chapter 2

Literature Survey

In this chapter, we explore several aspects of S2ST research including Speech Encoders, Speech to Unit Translation, End-to-end Speech to Speech Translation, Attention mechanisms for Simultaneous Translation, Simultaneous Speech to Text Translation and Automatic Speech Recognition.

2.1 Overview of Speech to Speech Translation

Speech-to-Speech Translation (S2ST) has become increasingly prominent within the realm of natural language processing, particularly with significant progress seen in end-to-end methodologies and text less contexts [1]. However, despite these advancements, there has been relatively limited exploration in adapting S2ST models for simultaneous applications within existing literature. This is notable, given the evident practical utility that simultaneous S2ST could provide, including applications like cross-lingual voice chat and live interpretation.

While offline Speech-to-Speech Translation (S2ST) has seen significant advancements, with complete input utterance translation prior to generating output, simultaneous S2ST presents distinct challenges. This arises from the necessity for real-time translation while continuously receiving input. Real-world applications, such as cross-lingual voice chat in international settings or live interpretation during multilingual events, demand efficient and accurate simultaneous S2ST systems.

However, research in this domain remains relatively underexplored compared to other facets of machine translation. This gap underscores the need for further investigation and the development of innovative approaches tailored to address the specific demands and hurdles of simultaneous S2ST. Hence, there is a pressing need to advance the state-of-the-art in simultaneous S2ST, and bridge the gap between offline S2ST and real-time applications. By leveraging the recent advancements in end-to-end and textless S2ST, and tailoring them to the unique demands of simultaneous scenarios, we can unlock the full potential of S2ST for practical and impactful use cases.

Although simultaneous speech to speech translation (S2ST) holds great potential for various applications, the development of direct end-to-end simultaneous S2ST systems remains an active research area, particularly for low-resource languages [2].

2.2 Speech Encoders

Self-supervised methods have gained traction for speech representation learning, yet they come with distinct challenges. This involves managing numerous sound elements within each input statement, absence of a lexicon for input sound elements during pre-training, and navigating through varying lengths of sound units without distinct segmentation.

In response to these challenges, Hsu et al. (2021) introduced the Hidden-Unit BERT (HuBERT) technique for self-supervised learning of speech representations. This method employs an offline clustering process to generate aligned target labels for a BERT-like prediction loss [3]. Notably, this loss function exclusively focuses on predicting masked tokens while disregarding non-masked ones. By concentrating the prediction loss solely on masked regions, the model is compelled to acquire a cohesive acoustic and language model across the uninterrupted input.

A crucial aspect of the HuBERT approach is its strong dependence on the consistency of the unsupervised clustering process rather than solely on the quality of the assigned cluster labels. Specifically, the HuBERT model begins with a basic k-means teacher consisting of 100 clusters and undergoes two rounds of clustering. Remarkably, it either matches or surpasses the performance of wav2vec 2.0 [4] on the Librispeech (960h) and Libri-light (60,000h) benchmarks across various fine-tuning subsets, including 10min, 1h, 10h, 100h, and 960h [5].

The HuBERT approach has demonstrated remarkable effectiveness in reducing Word Error Rate (WER) on the dev-other and test-other evaluation subsets. Notably, when employing a 1B parameter model, HuBERT achieves up to a 19% and 13% relative reduction in WER on these notably challenging subsets.

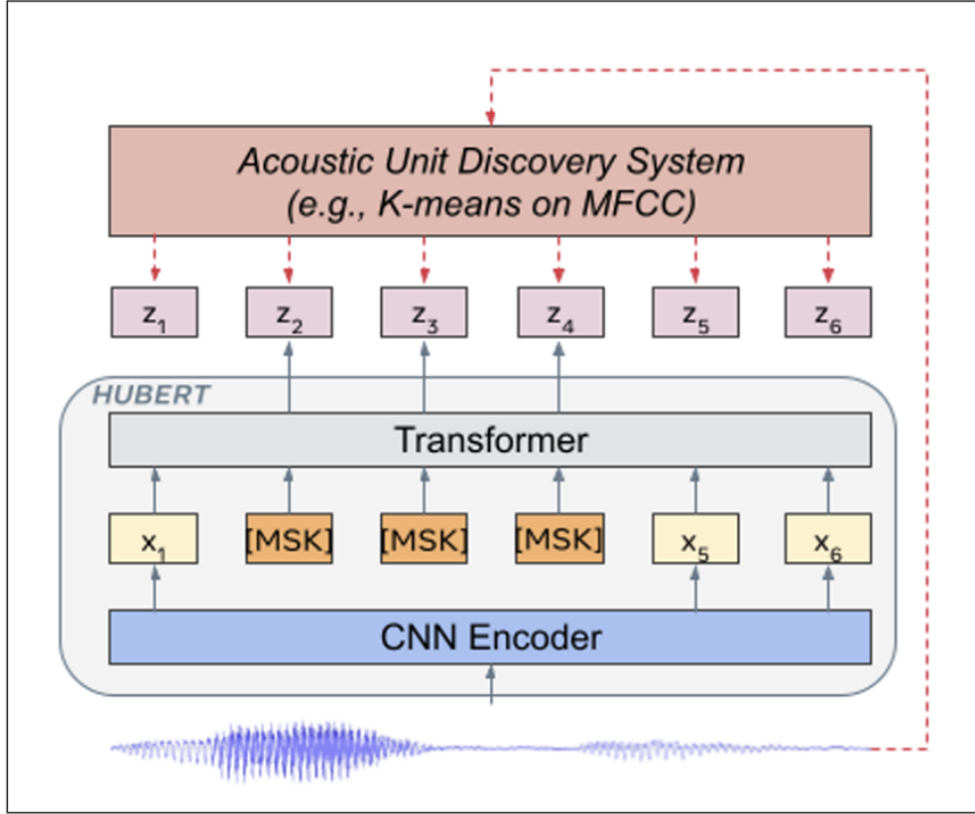


Figure 2.1: Prediction of hidden cluster assignments of masked frames.

2.3 Speech to Unit Translation

In recent years, there has been notable advancement in the domain of speech-to-speech translation, with numerous researchers exploring avenues to enhance the quality and efficiency of these systems. In this context [5] introduce a speech-to-unit translation (S2UT) model that harnesses self-supervised learning methodologies to address the constraints of existing systems. Significantly, they utilize HuBERT as a self-supervised discrete speech encoder for encoding target speech into discrete representations. Subsequently, they train a sequence-to-sequence speech-to-unit translation (S2UT) model to predict these representations.

In a related study, Lee et al. (2021b) extend the work outlined in Lee et al. (2021a) [20] by introducing a self-supervised unit-based speech normalization approach. This method refines a pre-trained speech encoder using paired audio samples from multiple speakers and a single reference speaker to mitigate accent variations while retaining lexical content. The researchers train their system on real-world speech-to-speech

translation (S2ST) data, encompassing diverse speakers and accents, to enhance the modeling of multi-speaker target speech.

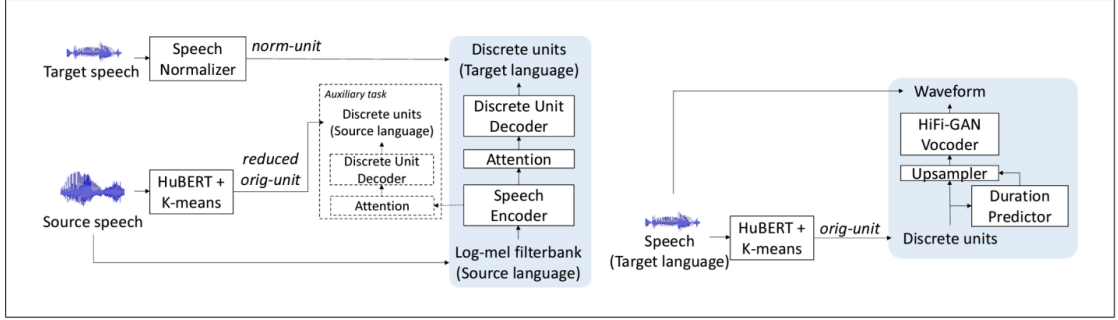


Figure 2.2: Illustration of the textless S2ST model.

2.4 End-to-End/Direct Speech to Speech Translation

Despite the inherent difficulties in attaining genuinely end-to-end speech-to-speech translation, a great deal of advancement has been made in this field. Jia et al. notably presented Translatotron, an end-to-end speech-to-speech translation system that employs a sequence-to-sequence model with attention, in 2019 and 2021. This network presents translated information with a unique canonical voice by using end-to-end training to learn how to map speech spectrograms from one language to target spectrograms in another. The researchers also show how well the system can mimic the original speaker's voice in the translated speech [7].

In contrast to alternative techniques, this method doesn't depend on intermediary representations like text or phonemes. Rather, it directly converts speech from one language to another, enabling the capture of richer information in both input and output signals, all while achieving lower latency. However, training end-to-end models proves more challenging due to the scarcity of training data and the intricacy of modeling the speech signal.

To assess the effectiveness of this method, experiments were carried out on two datasets for Spanish-to-English speech translation. The suggested model performs marginally less well than a baseline cascade that consists of a text-to-speech synthesis model after a direct speech-to-text translation model, according to the results. Despite this marginal discrepancy in performance, the findings illustrate the viability of this approach for tackling the demanding task of simultaneous speech-to-speech translation.

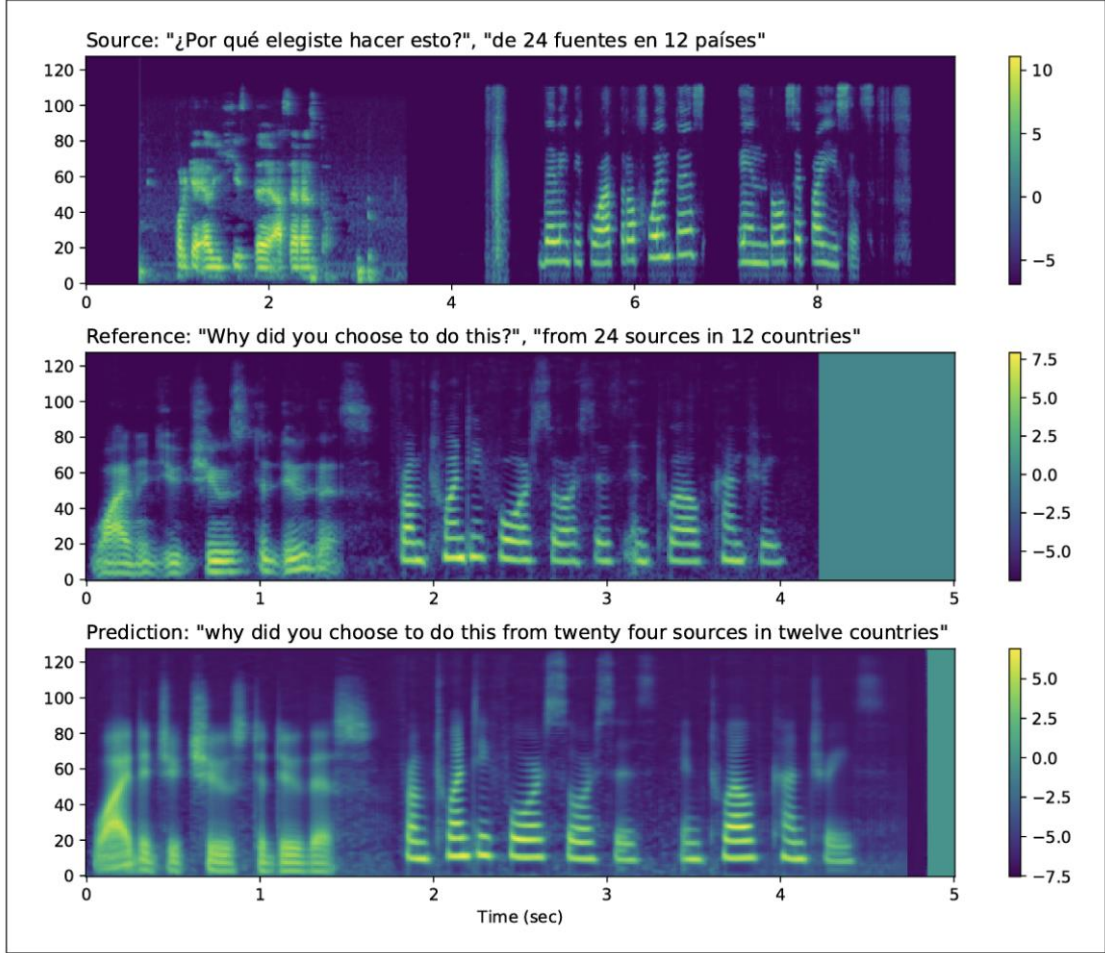


Figure 2.3: Sample mel-spectrograms

2.5 Attention Mechanisms

Attention-enhanced recurrent neural network (RNN) models have shown promise for a variety of sequence-to-sequence tasks in a variety of domains, including machine translation, speech recognition, and natural language processing [8]. The attention techniques enable the model to produce every element in the output sequence while dynamically allocating focus to different parts of the input sequence. Consequently, this capability leads to predictions that are remarkably expressive and contextually informed.

Despite their efficacy, traditional soft attention mechanisms are plagued by a drawback: they necessitate scanning the entire input sequence at every time step during inference, a process that can be computationally demanding and unfeasible for online scenarios. This stems from the mechanism's computation of a weighted sum of input sequence representations, wherein the weights are derived through a softmax operation

that encompasses the entirety of the input sequence. Consequently, the time complexity of soft attention mechanisms escalates quadratically with the input sequence's length, posing a potential bottleneck for real-time applications.

In the upcoming sections, we delve into different types of attention mechanisms pivotal to our comprehension of simultaneous speech-to-speech translation systems.

2.5.1. Monotonic Attention

To overcome the computational constraint discussed earlier, Raffel et al. (2017) introduce a novel approach that capitalizes on the observation that alignment between input and output sequence elements tends to be monotonic in numerous real-world scenarios. The primary focus of attention should be on adjacent elements within the input sequence as the model generates corresponding elements in the output sequence. Raffel et al. (2017) propose an end-to-end differentiable approach for learning monotonic alignments, building upon this insight. The online computation of attention in linear time during inference is made easier using this method [9].

This method empowers the model to calculate attention weights efficiently, relying on the alignment between input and output sequence elements in a monotonic fashion. It eliminates the necessity to scan the entire input sequence at each time step. This not only diminishes computational overhead but also renders the model suitable for online and real-time applications, where low-latency performance is paramount.

The proposed method has demonstrated its efficacy across numerous sequence-to-sequence tasks, ranging from sentence summarization and machine translation to online speech recognition. Experimental findings illustrate that this approach attains competitive performance levels when juxtaposed with existing sequence-to-sequence models. Additionally, it boasts the advantage of enabling online and linear-time computation of attention, enhancing its suitability for real-time applications.

2.5.2. Monotonic Chunkwise Attention

Chiu and Raffel (2017) propose an approach called Monotonic Chunkwise Attention (MoChA), which presents an adaptive solution to the limitations associated with soft attention mechanisms in online scenarios.

MoChA dynamically partitions the input sequence into smaller chunks, computing soft attention over these chunks instead of the entire input sequence. This chunkwise attention computation facilitates more efficient processing during both training and inference phases. MoChA offers a significant advantage with its compatibility with standard backpropagation, facilitating efficient model training. Moreover, during testing, MoChA enables online decoding in linear time, effectively addressing the quadratic time complexity concern inherent in traditional soft attention mechanisms. To elucidate the MoChA mechanism, Figure 2.4 provides a schematic representation of the model architecture.

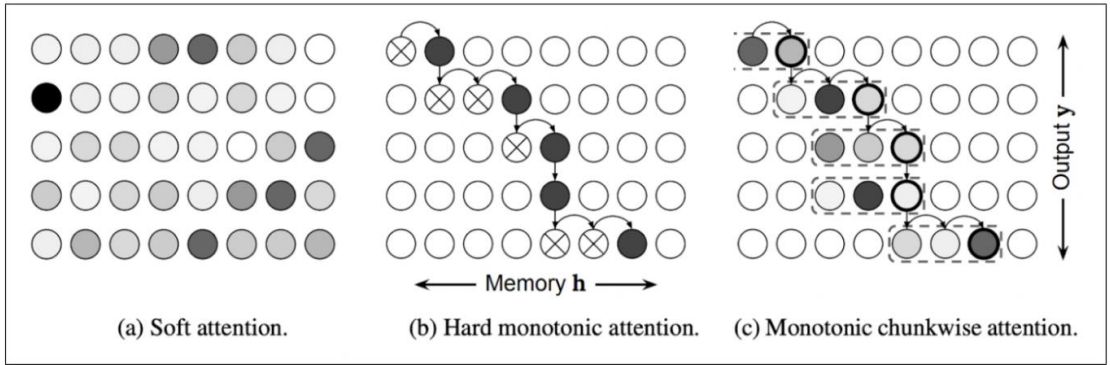


Figure 2.4: Schematics of the attention mechanisms

In this figure, nodes represent the possibility of the model attending to a memory entry at a given output time. The grey shading on each node represents the probability that the model assigns to each memory item at each output time in soft attention. Based on these probabilities, the context vector is calculated as the weighted average of the memory. Testing involves monotonic attention scanning memory entries (shown as black nodes) from left to right to decide whether to go on to the next entry or stop and attend. The attended memory entry receives the context vector, and the procedure continues from the previous attended entry at the next output time. MoChA employs a hard monotonic attention mechanism to ascertain the endpoint of the chunk to attend to, indicated by nodes with bold borders. The chunk boundaries, with a window size of 3, are delineated by dotted lines. Subsequently, the model conducts soft attention over the chunk, with the weighting depicted by the shade of gray on each node. The context vector is then computed as the weighted average of the chunk.

To assess the efficacy of MoChA, experiments were conducted on online speech recognition tasks. The findings indicate that MoChA achieves state-of-the-art performance at the time, comparable to a model employing an offline soft attention

mechanism. This underscores MoChA's ability to deliver highly accurate results while preserving online processing efficiency.

Furthermore, MoChA underwent evaluation on document summarization tasks, where monotonic alignments are not typically anticipated. Surprisingly, experiments unveiled that MoChA surpassed a baseline monotonic attention-based model, highlighting its potential to enhance performance in scenarios where monotonic alignments may not be prevalent [12].

2.5.3. Monotonic Infinite Lookback Attention

In 2019, Arivazhagan and colleagues present a novel simultaneous translation system that combines a neural machine translation (NMT) model with an adaptive schedule. This approach advances the idea of Monotonic Chunkwise Attention (MoChA) (Chiu and Raffel, 2017), which dynamically divides the input sequence into manageable pieces for the computation of soft attention. Monotonic Infinite Lookback (MILk) attention enables the model to conduct soft attention not only within the current chunk but also across the encoder states of the entire source sentence, granting the model an "infinite lookback" capability. Figure 2.5 depicts the schematic corresponding to the monotonic infinite lookback attention mechanism[15].

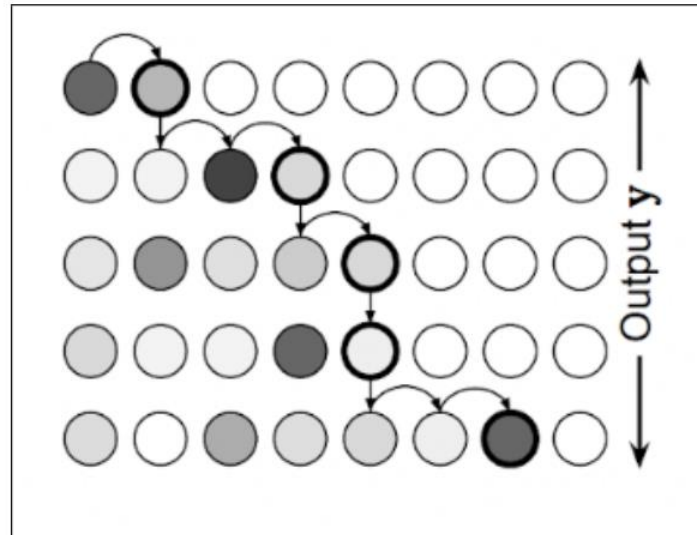


Figure 2.5: Schematics of the infinite lookback attention mechanism

The MILk attention mechanism in the suggested system consists of two parts: a soft attention head that goes all the way back to the beginning of the source sentence, and a hard, monotonic attention head that plans when to read the sentence. This distinctive fusion empowers the model to dynamically fine-tune its attention mechanism in response to the current input and generated translations, while also incorporating information from earlier segments of the source sentence. The adaptive schedule learned by MILk facilitates the model in achieving favorable trade-offs between latency and quality, surpassing the wait-k strategy across various latency values.

2.5.4. Monotonic Multihead Attention

The monotonic multihead attention mechanism, introduced by Ma et al. (2019), broadens the scope of the monotonic attention mechanism to incorporate multihead attention. Multihead attention allows the model to concentrate on various segments of the input sequence by dividing the input sequence into multiple subspaces, which are subsequently processed separately. Every head has the ability to distribute its attention to several locations, which allows it to process each incoming token while attending to prior states [16].

In contrast, the hard monotonic model, which utilizes only a single attention head, is limited in its capacity to attend to multiple positions simultaneously. Consequently, this can lead to the neglect of previous information at the attention layer as the model progresses to process new tokens. However, the introduction of multiple attention heads in the MMA model overcomes this constraint. It enables the model to dynamically adjust the pace of each head on-the-fly. While some heads focus on processing new inputs, others can remain fixed on the past to preserve the historical information of the source.

Figure 2.6 illustrates the schematic representation of the monotonic multihead attention mechanism introduced by Ma et al. (2019). The figure depicts various attention heads attending to distinct segments of the input sequence, where certain heads concentrate on past states while others prioritize the current state. Subsequently, the output of each head is concatenated and forwarded through a linear layer to generate the final attention output.

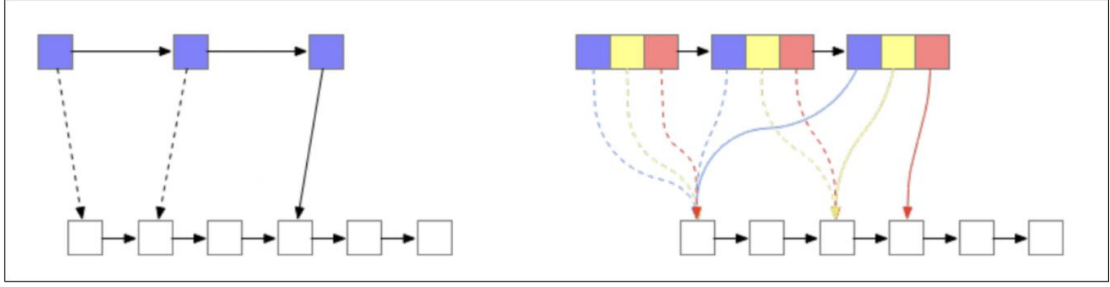


Figure 2.6: Monotonic Attention (Left) versus Monotonic Multihead Attention (Right)

2.6 Simultaneous Speech to Text Translation

Notable progress has been made in simultaneous text translation and end-to-end speech translation in the past few years, although both tasks are still considered independent. Nevertheless, very little research has been done on combining these responsibilities. Ma et al. (2020b) investigate how monotonic multihead attention and wait-k, two methods for simultaneous text translation, can be used to end-to-end simultaneous speech translation. Before choosing to READ or WRITE, they present a pre-decision module that helps organize encoder states into meaningful pieces.

The pre-decision module computes trigger probabilities (ptr) using the current encoder states and their associated features. A ptr value exceeding 0.5 signifies that a simultaneous decision should be made at that particular encoder state. Conversely, a ptr value less than or equal to 0.5 suggests that the model should continue reading new frames to accumulate additional information before making a decision. The architecture of Simul ST is depicted in Figure 2.7 [18].

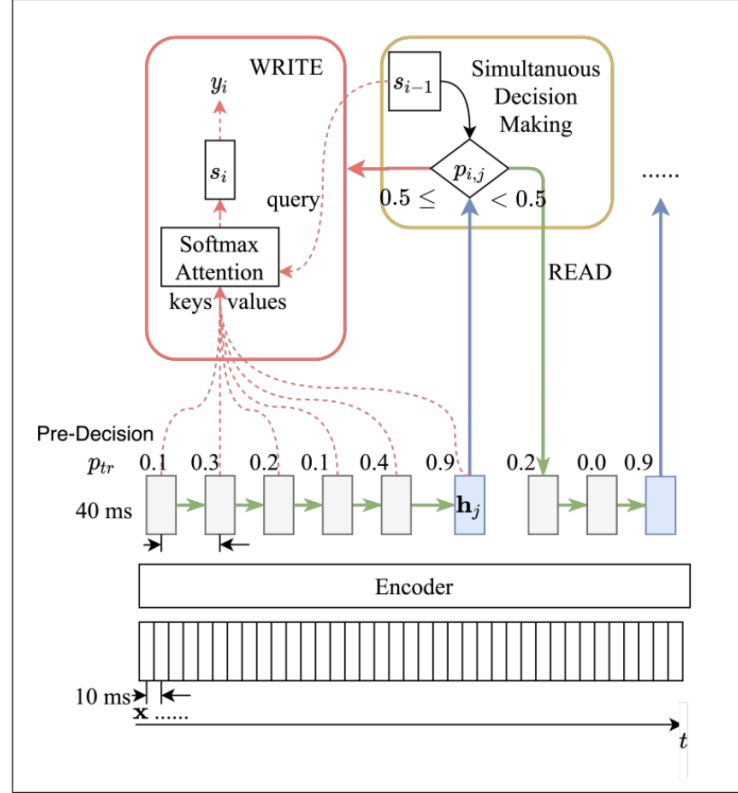


Figure 2.7: The Simul-ST architecture incorporates a pre-decision module. Blue states in the figure indicate the point at which the Simul-ST model initiates the simultaneous decision-making process.

Their study offers a comprehensive examination of the trade-offs between latency and translation quality by exploring the combination of fixed and flexible pre-decision strategies with fixed and flexible policies. To assess the effectiveness of their approach, they introduce a novel computation-aware latency metric derived from Average Lagging, which incorporates the computational cost of the translation process. This metric enables the quantification of the approach's efficiency in terms of computational resources and real-time translation performance.

2.7 Automatic Speech Recognition

Automatic Speech Recognition (ASR) encompasses research and technology aimed at creating systems capable of converting spoken language into written text automatically. It finds applications across various domains, including transcription services, voice assistants, speech-to-text translation, among others. ASR research entails the design and training of algorithms to accurately recognize and transcribe

speech, addressing challenges such as background noise, accents, dialects, variations in speaking rates, and language-specific traits.

The cutting-edge automated speech recognition (ASR) system Whisper was unveiled by Radford et al. (2022). It was trained on a large dataset that included 680,000 hours of multilingual and multitask supervised data that was obtained from the internet. Their findings highlight the ways in which this enormous and varied dataset has improved robustness against a range of obstacles, including accents, background noise, and technical terminology. Whisper is notable for its ability to translate voice into English and for its proficiency in transcribing speech in a variety of languages. Whisper's model and inference code are now accessible to the general public, providing a strong foundation for the creation of useful applications and additional developments in the field of robust speech processing[17].

Whisper's architecture is implemented as an encoder-decoder Transformer using a straightforward end-to-end concept. After being divided into 30-second segments, the audio input is transformed into a log-Mel spectrogram and then fed via an encoder. The matching text caption is generated by a decoder that has been taught to include unique tokens. These tokens are used to direct the single model while it does various tasks, such as language recognition, timestamping at the phrase level, transcription of multilingual speech, and English-to-language translation.

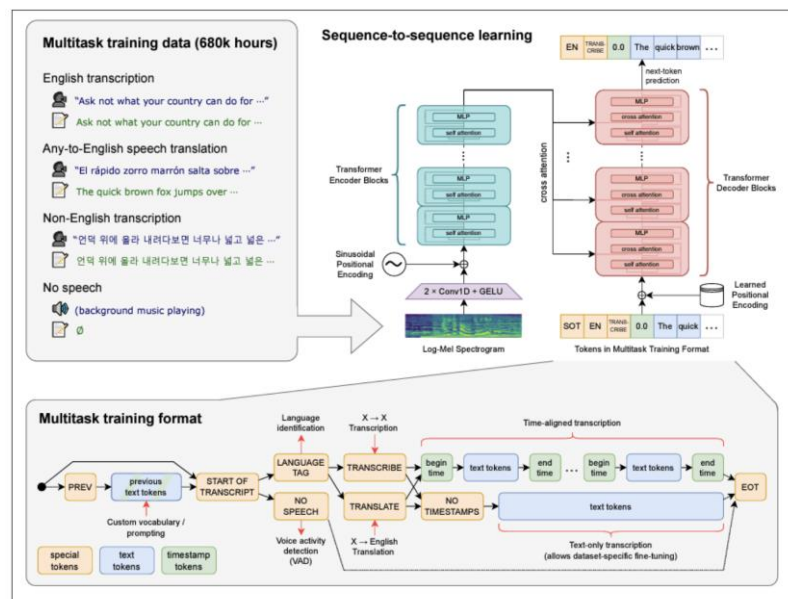


Figure 2.8: Sequence-to-sequence Transformer model

In comparison to other existing approaches that often rely on smaller and closely paired audio-text training datasets or unsupervised audio pretraining, Whisper distinguishes itself by being trained on a large and diverse dataset collected from the web [22]. Unlike models fine-tuned specifically for benchmark datasets such as LibriSpeech (Panayotov et al., 2015b), renowned for competitiveness in speech recognition, Whisper may not surpass them when both training and testing data are drawn from the same distribution. However, in assessing Whisper's zero-shot performance across diverse datasets, it demonstrates significantly enhanced robustness, exhibiting 50% fewer errors compared to those specialized models [23].

Approximately one third of the audio dataset used to train Whisper comprises non-English data, and the model is trained to transcribe the audio in the original language or translate it to English alternatively. It was observed that this approach is highly effective in learning speech to text translation, surpassing the supervised state-of-the-art (SOTA) performance on CoVoST2 [24] to English translation in a zero-shot setting. We use Whisper as the ASR model in our final experiment.

Chapter 3

Methodology

This chapter explores the creation of an advanced system designed for simultaneous speech-to-speech translation (S2ST), emphasizing its practical uses in live interpretation and cross-lingual voice chat. Our method overcomes the drawbacks of direct end-to-end techniques by introducing a two-phase procedure that consists of a text-to-speech (TTS) module that converts translated text into speech and a SimulST component that converts source speech to target text. This chapter examines our system's design and operation, highlighting its value in removing language barriers and promoting fluid communication in a variety of linguistic circumstances.

3.1 System Architecture

For simultaneous speech-to-speech translation (S2ST), emphasizing practical use cases such as cross-lingual voice chat and live interpretation, where our methodology overcomes the constraints of direct end-to-end approaches by employing a two-step process. This entails a SimulST component to convert source speech into target text, followed by a text-to-speech (TTS) module for generating target speech from the translated text. Figure 3.1 illustrates this cascaded architecture.

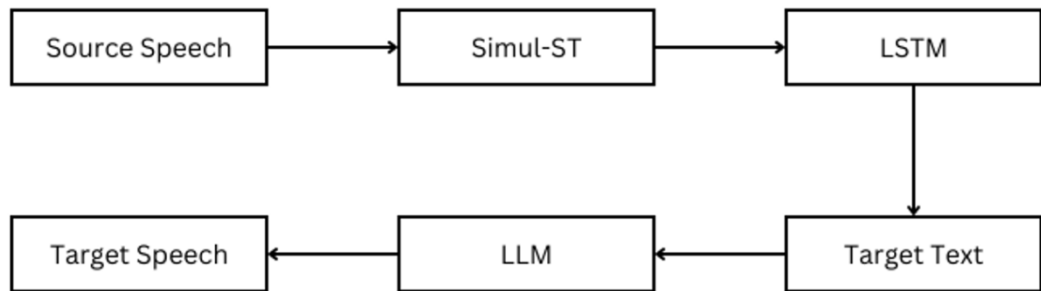


Figure 3.1: Schematic of Cascaded Simultaneous Speech to Speech Translation System

The adoption of this cascaded approach offers numerous advantages. Firstly, it enables us to capitalize on the advancements achieved in SimulST and TTS individually, facilitating the adaptation and optimization of each component for its specific task. This proves particularly advantageous for low-resource languages, where

training data for direct end-to-end S2ST may be scarce. Secondly, the cascaded architecture affords flexibility in selecting SimulST and TTS models, allowing us to independently choose the most effective models for each component, potentially enhancing the overall translation quality. Lastly, this cascaded system exhibits modularity, simplifying the process of updating or replacing individual components without disrupting the entire pipeline, thereby enhancing the system's adaptability and maintainability.

Our system offers broad language support, facilitating translation from diverse languages to English. A distinctive feature of our system lies in its capacity to dynamically regulate output latency via configurable parameters, inclusive of four policies governing the timing of output sequence generation.

3.2 Data Collection

The datasets have been used for training and evaluating various models, and have contributed significantly to the advancement of speech translation research.

3.2.1 CoVoST and CoVoST 2

The majority of language pairs with English as the source language, particular domains, or low-resource languages are the focus of the speech-to-text translation datasets now available. On the other hand, CoVoST, a multilingual speech-to-text translation corpus with 11 languages translated into English, is introduced by Wang et al. (2020a). Noteworthy for its diversity, CoVoST encompasses over 11,000 speakers and 60 accents, rendering it a valuable asset for multilingual ASR research. The authors detail the dataset's creation methodology, substantiate its data quality empirically, and present initial benchmarks, including what we believe to be the inaugural end-to-end many-to-one multilingual models for spoken language translation.

CoVoST V2, introduced by Wang et al. (2020b), marks a substantial enhancement of the CoVoST dataset tailored for multilingual speech-to-text translation (ST). This latest iteration presents a notably expanded multilingual ST dataset, facilitating translation from 21 languages into English and from English into 15 languages.

CoVoST V1 already boasted a remarkable collection of speech data, featuring over 11,000 speakers and 60 accents across languages such as French, German, Dutch, Russian, Spanish, Italian, Turkish, Persian, Swedish, Mongolian, and Chinese, totaling 708 hours of speech. In CoVoST V2, the dataset was expanded to include additional languages such as Welsh, Catalan, Slovenian, Estonian, Indonesian, Arabic, Tamil, Portuguese, Latvian, and Japanese, resulting in a substantial increase in the total amount of speech data available. CoVoST V2 encompasses a staggering 2,900 hours of speech, making it one of the most comprehensive multilingual speech-to-text translation datasets available for research and development purposes.

3.2.2 Common Voice Mozilla

Common Voice, developed by Mozilla, stands as an expansive open-source dataset crafted for training and assessing automatic speech recognition (ASR) models. With over 9,283 recorded hours and 7,335 validated hours across 60 languages, it ranks among the largest and most varied speech datasets available. What sets Common Voice apart is its inclusion of demographic metadata, encompassing factors like age, sex, and accent. This feature enables the analysis of model performance across diverse demographic profiles, fostering the development of voice technologies that are inclusive and effective for users from all backgrounds.

3.3 Algorithms

LSTM, a form of recurrent neural network (RNN), is crafted to capture extended dependencies within sequential data. It addresses the shortcomings of conventional RNNs by integrating memory cells and gating mechanisms. LSTMs find extensive application in diverse tasks like time series prediction, natural language processing, speech recognition, and others. Its essential elements, including the cell state, forget gate, input gate, and output gate, empower the model to retain and discard information across time steps. LSTMs have been instrumental in improving the performance of sequence prediction tasks and have become a fundamental building block in deep learning architectures.

Large Language Models (LLMs) are deep learning architectures that have been trained on large textual datasets to understand and generate text that is similar to that of a person. These models, which are usually based on transformer architectures, use self-

attention mechanisms to understand complex links in text over extended distances. Large-scale text corpora are used for rigorous pre-training of LLMs, which is followed by fine-tuning on specialised datasets for specific tasks. They do remarkably well on a variety of natural language processing tasks, including as text production, question answering, text translation, and text summarization. However, the development and deployment of LLMs raise ethical considerations regarding bias, misinformation, privacy, and potential misuse, requiring careful attention from researchers and practitioners.

LSTM and LLM algorithms have revolutionized the field of natural language processing and sequential data analysis, enabling significant advancements in various applications ranging from language understanding to text generation.

3.3.1 Long Sort Term Memory (LSTM) Algorithm

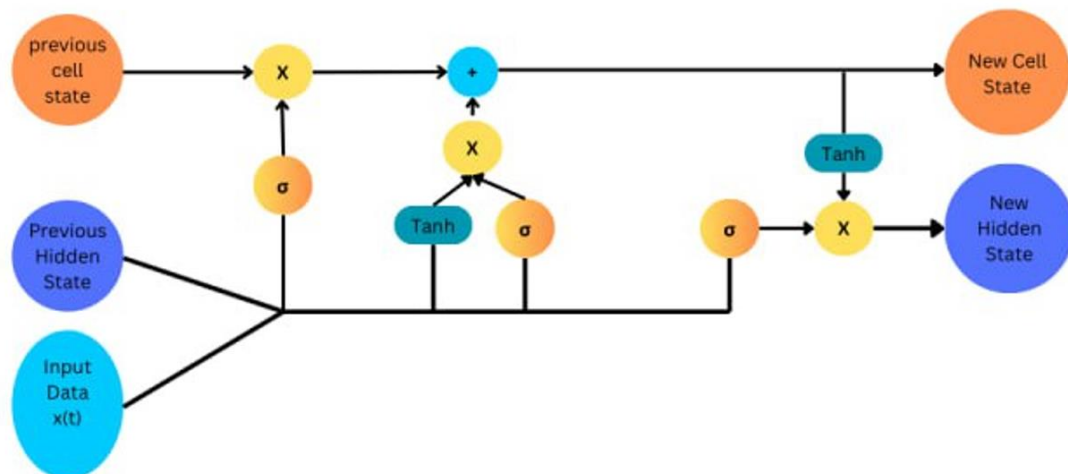


Figure 3.2 LSTM Algorithm flow chart

Recurrent neural network (RNN) architectures such as Long Short-Term Memory (LSTM) are ideal for learning patterns in sequential data, including text, audio, video, and time series data. In order to solve the vanishing gradient issue that plagues conventional RNNs and makes it challenging for RNNs to learn and remember information over lengthy sequences, Hochreiter & Schmidhuber invented LSTM networks in 1997.

1) Cell State: LSTM networks function as a conveyor belt along the entire sequence by maintaining a cell state. This cell state can have data added or withdrawn, enabling the

network to discover long-term dependencies. The capacity of LSTM to retain data over several time steps is an essential characteristic.

2) Gates: The forget gate, input gate, and output gate are the three gates that LSTM networks utilise to regulate the information flow. These gates are represented by sigmoid neural network layers, which determine what information should be remembered or forgotten based on input from the previous output and the current input.

- The Forget Gate determines which information from the previous cell state should be discarded.
- The input gate decides which new information should be stored in the cell state.
- The output gate, utilizing the current cell state, decides which information should be emitted as output.

3) Activation Functions: To control the information flow and enable the model to capture non-linear relationships in the data, LSTM units generally employ the tanh activation function. The gates also employ the sigmoid function to generate values between 0 and 1, which regulate the quantity of data that is allowed to pass through.

4) Backpropagation Through Time (BPTT): Backpropagation through time, a backpropagation variation created especially for sequential data, is used to train LSTM networks. Through the process of propagating errors back through the entire sequence, BPTT enables the network to update its parameters.

In several applications, such as speech recognition, machine translation, natural language processing (NLP), and more, LSTM networks have shown to be incredibly efficient. They have become a fundamental building block in deep learning architectures for sequential data processing.

3.3.2 Large Language Model (LLM)

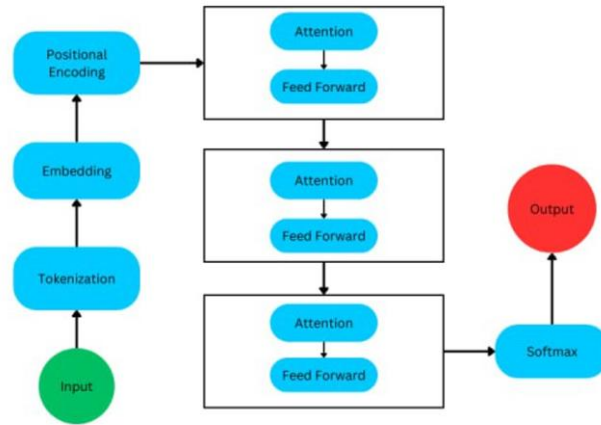


Figure 3.3 LLM Architecture

Large language models (LLMs) are a category of deep learning models designed to comprehend and generate text resembling human writing by leveraging patterns and structures identified in vast amounts of training data. These models find applications in various natural language processing (NLP) tasks, including text generation, question answering, translation, and summarization.

1) Architecture: Transformer architecture, first presented in the groundbreaking work "Attention is All You Need" by Vaswani et al. in 2017, is the foundation around which most LLMs are built. Transformers are able to capture long-range relationships more effectively than prior architectures like recurrent neural networks (RNNs) or convolutional neural networks (CNNs) because they rely on self-attention mechanisms to rate the relevance of different words in a phrase.

2) Training: Large volumes of text from sources including books, papers, internet, and social media—often hundreds of gigabytes or even terabytes—are needed to train LLMs. Usually, unsupervised learning methods like masked language modelling or autoregressive language modelling are used to train them.

3) Fine-tuning: To better suit certain NLP tasks, LLMs can be adjusted on smaller, task-specific datasets following their pre-training on a sizable corpus of text. Using

task-specific objective functions and supervised learning approaches, the model's parameters are updated during the fine-tuning process.

4) Applications: Large language models are useful in a wide range of fields. They are employed in text generation jobs including music composition, code generation, and article writing. They are also quite good at tasks like text categorization, named entity identification, sentiment analysis, and more.

5) Important ethical questions are brought up by the creation and application of large language models. These questions include those pertaining to bias, false information, privacy, and possible abuse. These issues are being actively addressed by researchers and practitioners in an effort to create plans for the responsible development and application of AI.

Two examples of large language models are OpenAI's GPT (Generative Pre-trained Transformer) series, such as GPT-2 and GPT-3, and Google's BERT (Bidirectional Encoder Representations from Transformers). These models continue to be the subject of continuing research and development efforts in the field of artificial intelligence, and they have considerably enhanced the state-of-the-art in natural language creation and understanding.

Chapter 4

Implementation

We conduct a thorough evaluation of our system, focusing on translating two linguistically diverse languages - Hindi, and Tamil - into English. We conduct our evaluations on a single CPU, leveraging OpenAI's Whisper for our offline ST model and the ElevenLabs API for our TTS model. The metrics we report are derived from a carefully selected subset of 75 examples from the CoVoST2 dev set. This subset is chosen to ensure comprehensive assessment across diverse language and speech segments, thereby facilitating robust performance evaluation.

4.1 Development Environment

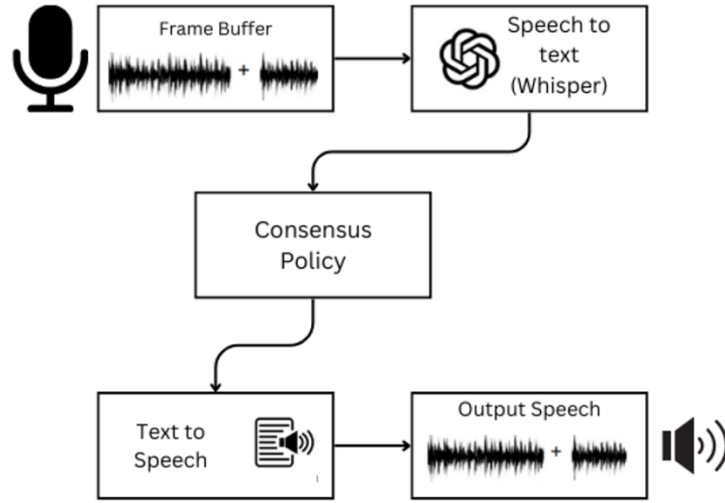


Figure 4.1 Proposed Cascaded S2ST System using OpenAI and ElevenLabs

First, speech segments are passed from the frame buffer to OpenAI's offline speech-to-text (ST) model, Whisper, in our cascaded SimulS2ST system. The translated text output is then generated based on the chosen policy for determining when to speak the output sequence. This approach ensures efficient processing of input speech frames by Whisper, resulting in real-time, accurate translations. The selected policy dictates when the translated text is spoken, ensuring natural and seamless speech synthesis. This

cascaded system facilitates effective and dynamic translation of speech input, catering to practical applications across various real-world scenarios requiring simultaneous translation.

Through the development of a resilient cascaded SimulS2ST system, our goal is to surmount existing research hurdles linked to direct end-to-end simultaneous S2ST, particularly concerning low-resource languages. Our envisioned approach holds promise in substantially propelling the domain of simultaneous S2ST forward, ushering in practical and efficient real-time translation systems applicable across diverse scenarios.

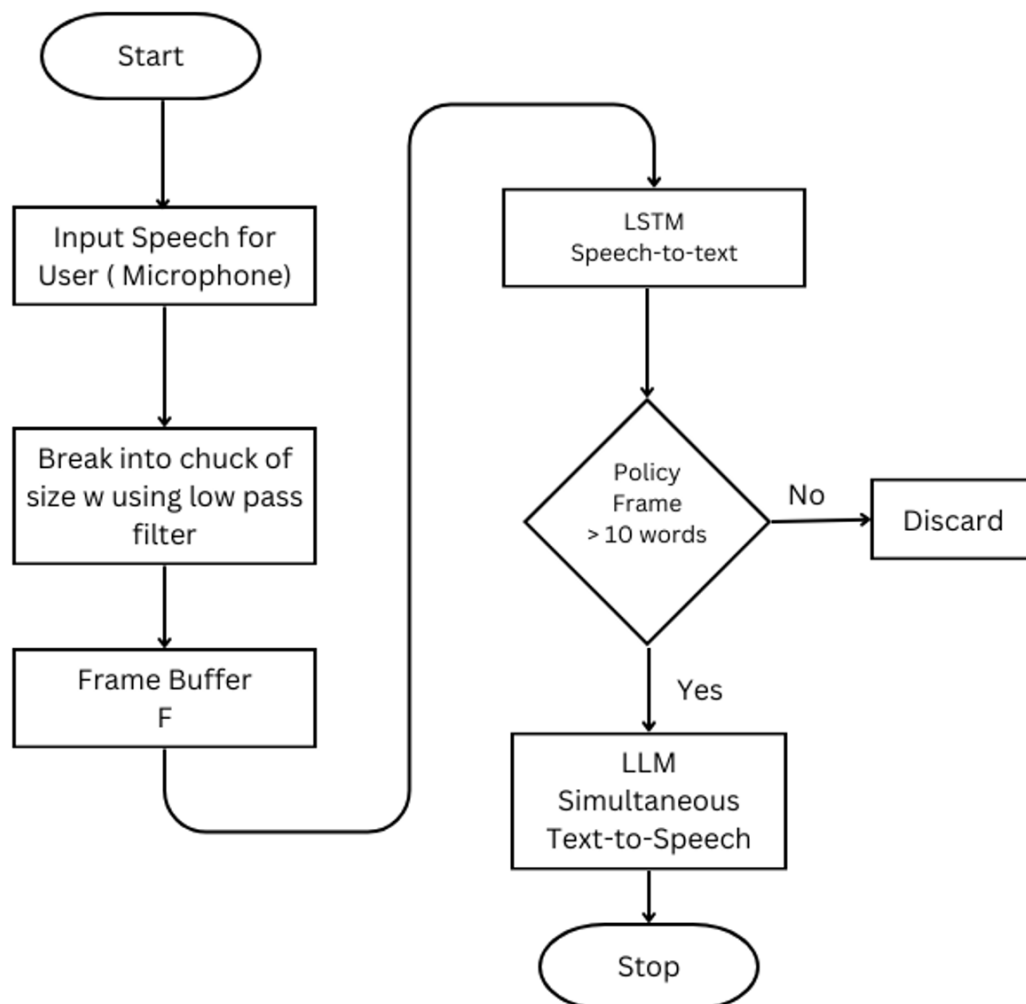


Figure 4.2: Development of Simultaneous Speech Translation system

In our assessment, we meticulously examine the interplay between latency and translation accuracy, employing four prototyped strategies to ascertain the timing of utterance delivery. Our objective is to achieve an optimal equilibrium between minimizing latency for real-time applications while upholding superior translation fidelity. By harnessing the capabilities of a readily available offline Speech-to-Text (ST) model in an online capacity, our aim is to propel SimulST research forward, offering valuable insights into the efficacy of various timing policies in real-time translation frameworks.

In our assessment, we meticulously examine the interplay between latency and translation accuracy, employing four prototyped strategies to ascertain the timing of utterance delivery. Our objective is to achieve an optimal equilibrium between minimizing latency for real-time applications while upholding superior translation fidelity. By harnessing the capabilities of a readily available offline Speech-to-Text (ST) model in an online capacity, our aim is to propel SimulST research forward, offering valuable insights into the efficacy of various timing policies in real-time translation frameworks.

4.1.1.SimulST Component: Real-time Speech-to-Text Translation with OpenAI Whisper

Inspired by Papi et al. (2022), our research takes a novel direction by employing an established offline Speech Translation (ST) model, namely OpenAI's Whisper (Radford et al., 2022), in an online setup to achieve precise translations with minimal latency. Rather than developing a distinct model tailored for Simultaneous Speech-to-Speech Translation (SimulST), we harness the functionalities of an off-the-shelf ST model to address real-time translation requirements.

We execute our method by dynamically querying the Whisper model during translation, enabling real-time generation of translations as input utterances arrive. This eliminates the necessity for preprocessing or buffering complete utterances before translation, effectively reducing system latency. Leveraging an established ST model trained on extensive data grants us the advantages of robustness and accuracy without the overhead of training a new model tailored for SimulST. In this specific toy experiment, our focus was on translating speech from Hindi to English text.

4.1.2 TTS Component: Chunkwise Text-to-Speech using ElevenLabs API

In the text-to-speech (TTS) component, various subcomponents collaborate to generate the intended output. This TTS module is structured into three interconnected parts, ensuring a cohesive and effective system.

The initial part of the TTS component involves the writer thread, tasked with emulating the streaming output of the speech-to-text translation module. This entails the creation of a pseudo thread that writes a string to a file, producing 5 words per second. By generating a continuous text stream in real time, this setup facilitates modularity and error containment between the SimulST and TTS components.

The subsequent component is the conversion thread, tasked with monitoring the file created by the writer thread for newly added content. Upon detecting new content, this thread utilizes the ElevenLabs API for speech conversion. The resulting speech is then saved as a WAV file, with each segment of processed text corresponding to a new file. Following WAV file creation, the conversion thread appends the file path to a queue for subsequent processing.

Lastly, the player thread continuously monitors the queue containing audio file paths. Upon detecting paths, it retrieves them one by one and plays the corresponding audio through the speaker. This sequential playback ensures the orderly and uninterrupted generation of audio output. The coordinated operation of these three components simulates a TTS module, showcasing its efficiency and effectiveness. For a visual demonstration of the process, a recording of this toy experiment is available at the provided link, offering insight into the functionality and utility of the TTS component in producing high-quality speech output from text.

4.1.3 Pipeline System Design

Figure 4.1 illustrates a schematic detailing the functioning of our system, which consists of a cascaded pipeline comprising the SimulST component followed by the TTS component. In essence, the system operates as follows: given an input sequence

denoted as S , denote the input sequence comprising N speech frames. We take a chunk of size w (where $w < N$) from the input in each iteration, and append it to the frame buffer F . Next, we present the frame buffer F and the spoken transcription as of right now T_{in} as input to the Speech to Text (ST) model, which generates the output text sequence T_{out} . The policy P receives this output text sequence T_{out} and decides whether or not to speak the generated sequence by the system.

This output text sequence T_{out} is then passed to the policy P , which determines whether or not the system should speak the generated sequence. If the policy P determines that the generated text sequence T_{out} should be spoken, we utilize the Text-to-Speech (TTS) model to convert T_{out} into speech, and append T_{out} to the current transcription T_{in} , after which we clear the frame buffer. On the other hand, if the policy P determines that the generated text sequence T_{out} should not be spoken, we discard T_{out} and wait for the next chunk of input speech to be retrieved from the input sequence S . This iterative process continues until the entire input sequence S has been processed.

To streamline execution and reduce output latency, we've designed our system to run the Simultaneous Speech to Text (SimulST) and Text-to-Speech (TTS) models on distinct threads. Additionally, for live microphone input, we've allocated a dedicated thread to the recorder, functioning concurrently with processing tasks. This threading strategy facilitates parallel execution of SimulST, TTS, and recording operations, thereby enhancing system efficiency.

The modular design of our pipeline enables users to seamlessly prototype and test various policies for deciding when to articulate the generated output utterances. This modular framework empowers users to assess the impact of different policies on system latency and accuracy, fostering iterative refinement and experimentation in Simultaneous Speech-to-Speech Translation (SimulS2ST) system development.

In our pipeline, the consensus mechanism is pivotal as it facilitates the integration of newly generated tokens into the Text-to-Speech (TTS) system for speech synthesis. To optimize pipeline latency, we've parallelized all pipeline stages, enabling the recorder, the Speech-to-Text (ST) system, and the consensus mechanism to operate concurrently without waiting for each other, thereby maximizing audio throughput.

By harnessing parallelization, the recorder operates independently of the ST system, and the ST system proceeds without waiting for the consensus mechanism. This design

fosters efficient and uninterrupted speech input processing, culminating in reduced system latency. Such a parallelized approach guarantees seamless and efficient pipeline operation, yielding quicker and more responsive real-time speech-to-speech translation.

4.2 BLEU Score evaluation

Using the SacreBLEU software, we calculate the BLEU score between the reference and the spoken transcript T_{in} in order to assess the accuracy of our system (Post, 2018). All the same, we do not apply ASR BLEU to the speech output in order to prevent possible imprecision due to cascading ASR failures. By leveraging the SacreBLEU package and concentrating on the spoken transcript and reference, our goal is to derive dependable and meaningful accuracy metrics reflecting the quality of our system's translations while sidestepping potential errors from cascading ASR output.

To evaluate the latency of our system, we employ the Computation-Aware Average Lagging (ALCA) metric, derived from Ma et al.'s (2020c) Average Lagging metric. ALCA considers the computational expense involved in producing an output, a pertinent factor for our system as it incurs computational costs during output speech generation and thread context switching. By integrating computational cost into our latency metric, we aim for a more holistic and precise evaluation of the system's overall performance, encompassing translation quality and computational efficiency.

4.3 Policy Comparisons

In our evaluation, we compared four policies dictating when to deliver output speech. Two serve as baselines: the Greedy Policy, setting $P(T_{out}) = \text{True}$, and the Offline Policy, with $P(T_{out}) = \text{False}$. These extremes provide insights into the highest and lower bounds of our system's performance across these metrics, representing the trade-off spectrum between translation quality and delay.

Following the establishment of baseline policies, the Confidence-Aware Policy (CAP) is introduced. CAP activates output speech when the average probability of the sequence generated by the ST model surpasses a predefined threshold, utilizing the Whisper field labeled as `no_speech_prob`. This methodology demonstrated superior

empirical outcomes when compared to utilizing the returned `avg_logprob`, consequently, it is incorporated into our evaluation procedure.

Finally, the Consensus Policy (CP) is introduced, triggering output speech only when the ratio of length to edit distance between the current transcript and the previous transcript falls below a certain threshold α . We demonstrate the flexibility of our system to varied latency regimes by running both the Consensus Policy and the Confidence-Aware Policy with two distinct parameter configurations.

Chapter 5

Results and Discussion

The performance and efficacy of our speech-to-speech translation system, we conducted comprehensive simulations across diverse scenarios and languages. The simulations aimed to replicate real-world usage conditions, allowing us to assess the system's capabilities and generate meaningful results.

5.1 Simulation Output:

We've engineered a system that merges translation and text-to-speech (TTS) modules, facilitating simultaneous speech-to-speech translation. This technology swiftly translates spoken language into text in the target language and seamlessly converts it to speech, enabling real-time communication.

5.1.1. BLEU Score

Table 5.1 presents the evaluation results, which indicate that languages that are structurally comparable to English generally show reduced latency for all policies. Particularly, we note significant latency reductions for Hindi across most policies, with fewer instances observed for Tamil. This underscores the importance of tailoring model input parameters to individual languages, underscoring the necessity for language-specific optimization strategies.

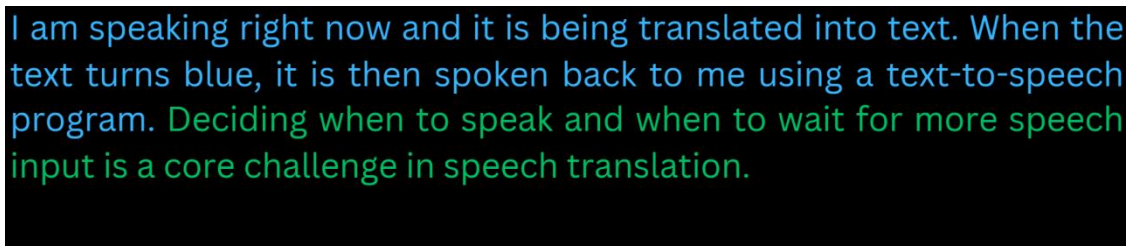
Table 5.1 BLEU Score Comparison

Window size(t)	Hindi 1s	English 3s	Tamil 1s	English 3s
CAP ($\gamma=0.75$)	20.65(7.2)	21.3(8.6)	38.2(6.4)	41.25(7.6)
CAP($\gamma=0.5$)	14.1(4.9)	18.9(6.7)	25.75(3.5)	28.2(5.4)
CP($\alpha= 0.75$)	16.2(6.8)	20.3(9.6)	30.1(4.6)	40.4(7.9)
CP($\alpha=0.5$)	15.4(4.8)	20.4(8.8)	25.3(4.1)	35.55(6.1)
Greedy(waitK)	5.65(3.2)	10.2(4.9)	8.4(2.5)	15.3(4.1)
Offline Policy	21.7(9.5)	21.8(9.5)	42.9(9.6)	42.9(9.6)

Bold text indicates the ideal latency-quality trade-off. English-like structural languages show superior trade-off. In comparison to the Greedy strategy, Hindi and Tamil BLEU scores approach offline levels with a negligible delay increase.

Additionally, our evaluation reveals significant enhancements in both translation quality and latency for specific languages, outperforming the Greedy baseline while maintaining a comparable translation quality to the offline setting. For instance, when translating Hindi to English using the Confidence-Aware Policy with a threshold of $\gamma = 0.5$, we achieve a remarkable 17-point increase in BLEU score, with only a minor sacrifice of 1 second in average lagging. Similarly, in the case of Tamil, employing the Consensus Policy with a threshold of $\alpha = 0.75$ results in a nearly 15-point increase in BLEU score, with a modest increase of 1.6 seconds in latency. These high BLEU scores indicate the promising practical utility of our system in real-world scenarios.

The system demonstrated accurate translations across the selected languages, maintaining a high level of fidelity to the source content. The system successfully handled dynamic language switching, adapting to changes in language input without compromising translation quality. Robustness testing showed the system's resilience to environmental noise and disturbances, ensuring reliable performance in diverse settings. The system exhibited low latency, providing real-time translations and contributing to a seamless user experience.

A screenshot of a text-based interface with a black background. The text is displayed in two colors: blue and green. The first sentence, "I am speaking right now and it is being translated into text. When the text turns blue, it is then spoken back to me using a text-to-speech program.", is in blue. The second sentence, "Deciding when to speak and when to wait for more speech input is a core challenge in speech translation.", is in green. This visual representation likely indicates different states or processing stages of the translation system.

I am speaking right now and it is being translated into text. When the text turns blue, it is then spoken back to me using a text-to-speech program. Deciding when to speak and when to wait for more speech input is a core challenge in speech translation.

Figure 5.1: Visual Output of Cascaded S2ST system using OpenAI

5.1.2. Word Accuracy

Evaluating word accuracy in a Simultaneous speech-to-speech translation system is essential for assessing the fidelity of the translated output compared to the original speech input. Transcribe the reference speech (original user input) into text using a reliable automatic speech recognition (ASR) system or manual transcription. Apply the speech-to-speech translation system to translate the transcribed reference speech into the target language. Obtain the reference transcription (original speech in text form) and the system-generated transcription (translated speech in text form). Tokenize both transcriptions into individual words. You can use libraries like NLTK or SpaCy for this task.

The numerator represents the count of correctly aligned words, and the denominator is the total number of words in the reference transcription.

We used NLTK for our project evaluation and the results are :

```
[nltk_data] Downloading package punkt to  
[nltk_data] C:\Users\Gowthami\AppData\Roaming\nltk_data...  
[nltk_data] Package punkt is already up-to-date!  
Word Accuracy: 83.33%  
>>> |
```

Figure 5.2 : Results of Word accuracy

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

The project "Speech to Speech Translation Using Machine Learning Algorithms" represents a significant step forward in addressing the complex challenges associated with real-time, simultaneous speech-to-speech translation. The research and implementation have provided valuable insights into the delicate balance between latency and translation quality, offering a nuanced understanding of the trade-offs involved. Simultaneous speech to speech translation (SimulS2ST) is a burgeoning field with promising practical applications in communication enhancement. To facilitate further research and development in this area, we've devised a flexible baseline system enabling users to fine-tune policy parameters, thus influencing the balance between translation quality and latency. Our evaluations demonstrate that these parameters yield accuracy comparable to offline models while notably enhancing latency performance. We anticipate that our system will prove invaluable to industry practitioners and researchers alike, fostering the prototyping and advancement of future SimulS2ST systems and contributing to the ongoing evolution of this field.

6.2 Future Scope

Investigating the properties of hidden units in neural networks is a fascinating field of study. Using inflected speech data to train the Hidden Unit encoder network and evaluate its capacity to capture these qualities is one possible avenue. In order to use this method, artificial training datasets would need to be created by adding variances in speech inflection to pertinent tokens of interest.

Offering a thorough comparison between end-to-end simultaneous speech to speech and the method that use text as an intermediary representation is another exciting avenue. This would make it easier to comprehend the trade-offs between these two strategies and recognise the advantages and disadvantages of each. The project paves the way for future research and development to further propel the field of simultaneous speech-to-speech translation forward. Continuous refinement of language models to

capture evolving linguistic nuances, idioms, and cultural variations, ensuring the system remains at the forefront of translation accuracy.

References

- [1] Learning When to Speak: Latency and Quality Trade-offs for Simultaneous Speech-to-Speech Translation with Offline Models Liam Dugan^{1,2}, Anshul Wadhawan¹, Kyle Spence², Chris Callison-Burch¹, Morgan McGuire², Victor Zordan² ¹University of Pennsylvania, ²Roblo
- [2] Y. Jia, M. T. Ramanovich, T. Remez, and R. Pomerantz, “Translatotron 2: High-quality direct speech-to-speech translation with voice preservation,” in International Conference on Machine Learning. PMLR, 2022, pp. 10 120–10 134.
- [3] A. Lee, H. Gong, P.-A. Duquenne, H. Schwenk, P.-J. Chen, C. Wang, S. Popuri, Y. Adi, J. Pino, J. Gu, and W.-N. Hsu, “Textless speech-to-speech translation on real data,” 2022
- [4] H. Inaguma, S. Popuri, I. Kulikov, P.-J. Chen, C. Wang, Y.-A. Chung, Y. Tang, A. Lee, S. Watanabe, and J. Pino, “Unity: Twopass direct speech-to-speech translation with discrete units,” 2022
- [5] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” 2022.
- [6] ElevenLabs, “Elevenlabs API,” <https://api.elevenlabs.io/docs>, 2023, accessed: 2023-04-10.
- [7] Direct speech-to-speech translation with a sequence-to-sequence model Ye Jia* , Ron J. Weiss* , Fadi Biadsy, Wolfgang Macherey, Melvin Johnson, Zhifeng Chen, Yonghui Wu
- [8] ESPnet-ST: All-in-One Speech Translation Toolkit Hirofumi Inaguma ¹ Shun Kiyono ² Kevin Duh ³ Shigeki Karita ⁴ Nelson Yalta ⁵ Tomoki Hayashi ⁶ , ⁷ Shinji Watanabe

[9] Multimodal and Multilingual Embeddings for Large-Scale Speech Mining Paul-Ambroise Duquenne ,Hongyu Gong ,Holger Schwenk

[10] CVSS Corpus and Massively Multilingual Speech-to-Speech Translation Ye Jia, Michelle Tadmor Ramanovich, Quan Wang, Heiga Zen

[11] LibriS2S: A German-English Speech-to-Speech Translation Corpus Pedro Jeuris and Jan Niehues Department of Data Science and Knowledge Engineering, Maastricht University Maastricht, The Netherlands

[12] Leveraging Pseudo-labeled Data to Improve Direct Speech-to-Speech Translation Qianqian Dong^{1,*} , Fengpeng Yue^{2,*,†} , Tom Ko¹ , Mingxuan Wang¹ , Qibing Bai^{2,†} , Yu Zhang²,

[13] JOINT PRE-TRAINING WITH SPEECH AND BILINGUAL TEXT FOR DIRECT SPEECH TO SPEECH TRANSLATION Kun Wei^{1,†} , Long Zhou² , Ziqiang Zhang² , Liping Chen² , Shujie Liu² , Lei He² , Jinyu Li² , Furu Wei²

[14] UnitY: Two-pass Direct Speech-to-speech Translation with Discrete Units Hirofumi Inaguma[♡], Sravya Popuri[♡], Ilia Kulikov[♡], Peng-Jen Chen[♡], Changhan Wang[♡], Yu-An Chung[♡], Yun Tang[♡], Ann Lee[♡], Shinji Watanabe[♣], Juan Pino[♡]

[15] BLASER: A Text-Free Speech-to-Speech Translation Evaluation Metric Mingda Chen, Paul-Ambroise Duquenne, Pierre Andrews, Justine Kao, Alexandre Mourachko, Holger Schwenk^{*} , Marta R. Costa-jussà^{*}

[16] Unit-based Speech-to-Speech Translation Without Parallel Data Anuj Diwan Anirudh Srinivasan David Harwath Eunsol Choi

[17] Learning When to Speak: Latency and Quality Trade-offs for Simultaneous Speech-to-Speech Translation with Offline Models Liam Dugan¹², Anshul Wadhawan¹ , Kyle Spence² , Chris Callison-Burch¹ , Morgan McGuire² , Victor Zordan²

- [18] C. Wang, A. Wu, and J. Pino, “Covost 2 and massively multilingual speech-to-text translation,” 2020.
- [19] M. Post, “A call for clarity in reporting BLEU scores,” in Proceedings of the Third Conference on Machine Translation: Research Papers. Belgium, Brussels: Association for Computational Linguistics, Oct. 2018, pp. 186–191. [Online]. Available: <https://www.aclweb.org/anthology/W18-6319>
- [20] X. Ma, J. Pino, and P. Koehn, “Simulmt to simulst: Adapting simultaneous text translation to end-to-end simultaneous speech translation,” arXiv preprint arXiv:2011.02048, 2020.
- [21] Rosana Ardila, Megan Branson, Kelly Davis, Michael Henretty, Michael Kohler, Josh Meyer, Reuben Morais, Lindsay Saunders, Francis M. Tyers, and Gregor Weber. Common voice: A massively multilingual speech corpus, 2020.
- [22] Naveen Arivazhagan, Colin Cherry, Wolfgang Macherey, Chung-Cheng Chiu, Semih Yavuz, Ruoming Pang, Wei Li, and Colin Raffel. Monotonic infinite lookback attention for simultaneous machine translation, 2019.
- [23] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations, 2020a. Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations, 2020b.
- [24] William Chan, Daniel Park, Chris Lee, Yu Zhang, Quoc Le, and Mohammad Norouzi. Speechstew: Simply mix all available speech recognition data to train one large neural network, 2021.
- [25] Matt Post. A call for clarity in reporting BLEU scores. In Proceedings of the Third Conference on Machine Translation: Research Papers, pages 186–191, Belgium, Brussels, October 2018. Association for Computational Linguistics.

- [26] Wu. BigSSL: Exploring the frontier of large-scale semi-supervised learning for automatic speech recognition. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1519–1532, oct 2022. doi: 10.1109/jstsp.2022.3182537.
- [27] Chung-Cheng Chiu and Colin Raffel. Monotonic chunkwise attention, 2017. URL <https://arxiv.org/abs/1712.05382>.
- [28] ElevenLabs. Elevenlabs API. <https://api.elevenlabs.io/docs>, 2023.